

An 8-Bit 2-GSPS Time-Interleaved SAR ADC using 45-nm CMOS

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Abstract—This paper presents a design pipeline and simulation-based verification for an 8-bit, 2-GSPS analog-to-digital converter implemented using 45-nm CMOS technology and a supply of 1.1 V. The successive approximation register (SAR) topology was chosen above other topologies due to its scalability at low GHz frequencies. The design includes nine SAR sub-channels that are time-interleaved to achieve a target of 2-GSPS sample rate. Each sub-channel contains a bootstrapped sample-and-hold switch, a PMOS-input StrongArm comparator, a segmented capacitive DAC with a 20 fF unit capacitor, and a FSM-based SAR control logic block. Transistor sizing was achieved through iterative simulations to find working parameters. Simulation results at a low frequency (5 MHz), near Nyquist frequency (990 MHz), and near sampling frequency (1990 MHz) resulted in SFDR values of 46.61, 42.23, and 32.51 dBc, and ENOBs of 5.64, 5.58, and 4.18 bits respectively. The average FOM across the three test conditions is 46.39 fJ/step.

I. INTRODUCTION

High-speed ADCs are essential for modern computational tasks especially in software-defined radio and radar systems [1]. This project presents a design for an 8-bit Nyquist ADC operating at 2 GS/s with the lowest achievable figure of merit (FOM) designed using the CMOS 45 nm predictive transistor model. The design has a V_{dd} of 1.1 V and a full peak-to-peak voltage swing of 0.5 V.

Architecture selection:

Three different ADC topologies were initially considered: Flash, pipelined, and SAR. The SAR topology was ultimately chosen due to its simple and efficient design, requiring only one comparator and one DAC per channel. This results in low power consumption and thus increases scalability, making it an industry standard design for low GHz processes processes and a good candidate for minimizing FOM [2]. The design specifications for this project are summarized in Table I and the block diagram for the full time-interleaved ADC system is shown in Figure 1

II. ANALYSIS

The initial circuit design was implemented in Cadence with L = 50nm and W = 1μm chosen as initial design sizings based on prior familiarity with similar 45 nm CMOS designs, providing a reasonable starting point for convergence. Transistor sizing, capacitor values, and operating points were then iteratively

TABLE I
DESIGN SPECIFICATIONS

Parameter	Value
Technology	CMOS 45-nm predictive
Supply Voltage	1.1 V
Architecture	9× Time-Interleaved SAR
Resolution	8 bits
Sample Rate	2 GSPS
Input Range	0.05–0.55 V ($V_{pp} = 0.5$ V)
Common-Mode	0.3 V
Unit Capacitance	20 fF
Clock Rise/Fall	100 ps (ideal)

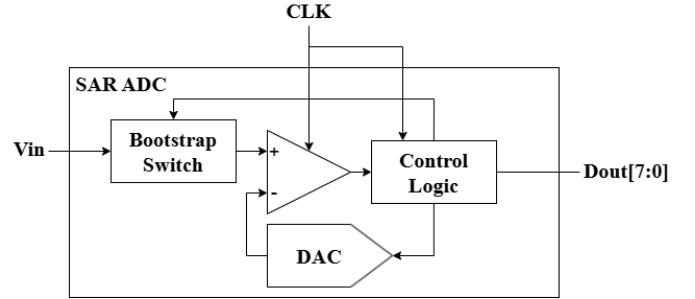


Fig. 1. Block diagram of single channel SAR ADC.

redefined through SPICE-based simulations in Cadence to find the working design parameter values.

A. Capacitor Sizing

For an N -bit ADC, the ideal quantization SNR floor is [3]:

$$\text{SNR}_{\text{ideal}} = 6.02N + 1.76 \approx 49.9 \text{ dB} \quad (N = 8).$$

The LSB voltage for $V_{pp} = 0.5$ V is $V_{\text{LSB}} = 0.5/256 \approx 1.95$ mV. The thermal noise on the sampling capacitor is $v_n = \sqrt{kT/C}$ [4]. To keep sampling noise below one quantization level ($v_n^2 < V_{\text{LSB}}^2/12$), the theoretical minimum capacitance is [4]:

$$C_{\text{min}} = \frac{12kT}{V_{\text{LSB}}^2} = \frac{12 \times 1.38 \times 10^{-23} \times 300}{(1.95 \times 10^{-3})^2} \approx 13 \text{ fF}.$$

This minimum bound was used for selecting the starting point of $C_{\text{unit}} = 20$ fF.

B. Transistor Mismatch

Design requirements specify that the matching property for the 45-nm CMOS technology is assumed $A_{vt} = 5 \text{ mV} \cdot \mu\text{m}$, $A_C = 0.3\% \sqrt{\text{fF}}$, and $A_R = 0.5\% \mu\text{m}$, giving the following formulas:

$$\sigma_{V_t} = \frac{A_{vt}}{\sqrt{WL}}, \quad \frac{\sigma_C}{C} = \frac{A_C}{\sqrt{C}}.$$

These formulas provided a basis for the initial sizing targets for the comparator input pair and CDAC unit capacitor. For the 20-fF unit cap, $\frac{\sigma_C}{C} = \frac{A_C}{\sqrt{C}} = \frac{0.3\%}{\sqrt{20 \text{ fF}}} = \frac{0.3\%}{4.47} \approx 0.067\%$, which is below the linearity requirement of $\frac{1}{2^8} = \frac{1}{256} \approx 0.39\%$. As mentioned previously, the final device dimensions were determined through Cadence parametric sweeps of transistor widths and lengths to optimize for speed, gain, and offset given the 1.1 V supply. Finite bandwidth of the bootstrapped switch constrains the $R_{\text{on}} \cdot C_{\text{load}}$ time constant to ensure settling within the 500 ps sampling slot, and static comparator offset due to input pair mismatch ($\sigma_{V_t} = A_{vt}/\sqrt{WL}$) was managed through transistor sizing to keep the 3σ offset below $V_{\text{LSB}}/2$. This residual offset contributes to the observed SINAD floor in the results.

III. CIRCUIT IMPLEMENTATION

A. Track-and-Hold: Bootstrapped Switch

For the track-and-hold (or sample-and-hold), a bootstrap switch topology was chosen referencing the design from [2] (see Appendix A for the schematic and symbol). The purpose of the bootstrap switch is to make sure the input is accurately sampled on the sampling capacitor. This capacitor holds the sampled value during SAR conversion while the comparator finds the digital bit values. A bootstrap was chosen rather than a plain NMOS gate or some other simpler configuration because of the effects of R_{on} at higher MHz frequencies. For the 990 MHz (near Nyquist of the sub-channel) test or above, a plain NMOS switch exhibits input-dependent R_{on} . The dependence of the on resistance on the input causes nonlinear distortion that surpasses the 8-bit noise floor, reducing ENOB. The bootstrapped switch eliminates this by holding $V_{\text{GS}} \approx V_{\text{DD}}$ independent of the input level. This results in near constant R_{on} without a boosted supply. Additionally, the settling time for this configuration fits in the 500 ps time-interleaved slot, making the Nyquist rate sampling possible, preserving ENOB near the Nyquist frequency.

B. Comparator: PMOS-Input StrongArm Latch

A StrongArm dynamic latch was used as the comparator (see Appendix A for the schematic and symbol). The StrongArm topology, influenced largely by [2], was optimal for the design as it naturally produces rail-to-rail outputs that drive the SAR logic directly. A PMOS input differential pair was chosen because $V_{\text{cm}} = 0.3 \text{ V}$ does not provide enough voltage headroom to bias an NMOS input pair in saturation for a 1.1 V supply since the minimum common-mode voltage is approximately $V_{\text{CM}} \approx V_{\text{TH}} + V_{\text{OV}} > 0.3 \text{ V}$. This makes NMOS operation impossible since the transistors would not

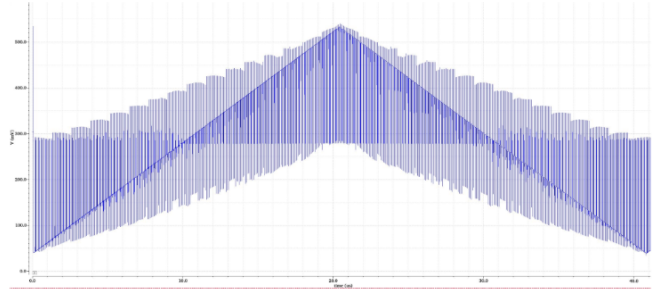


Fig. 2. Ramp simulation of linear behavior across 256 output codes.

stay in saturation and thus not operate correctly as an input pair. The input common-mode was adjusted from the originally intended 0.3–0.8 V range to 0.05–0.55 V ($V_{\text{cm}} = 0.3 \text{ V}$) to meet the headroom requirements of the PMOS comparator and bootstrapped switch at the 1.1 V supply. Finally, a SR latch was added to the StrongArm design to make sure that the result of the comparator output (the sampled digital value) is held and not affected by the comparator's precharge stage.

C. DAC: Segmented Capacitive DAC

A segmented CDAC with a 4-bit main array and 4-bit sub-array coupled through $C_C = (16/15) C_{\text{unit}}$ is used (see Appendix A for the schematic and symbol). This configuration was chosen because the segmented structure reduces total capacitance by $\approx 32\times$ vs. a full binary-weighted 8-bit array. This results in lower settling time and switching energy. Control signals $\text{sp}[k]$, $\text{sn}[k]$, and s_en connect each cap plate to V_{refp} , V_{refn} , or V_{cm} , respectively. A ramp simulation confirms monotonic, linear behavior across all 256 output codes (see Figure 2).

Area breakdown: Area was estimated using a metal capacitor density of $2 \text{ fF}/\mu\text{m}^2$ for capacitors, and transistor area computed from $W \times L$ dimensions in the 45-nm process. The area estimate for each block in a single channel is summarized in Table II. The dominant contributor is the DAC, whose capacitor array is responsible for the majority of area. Summing all blocks across all 9 interleaved channels and the single shared clock generation and MUX logic yields a total estimated area of $4446.7 \mu\text{m}^2 \approx 0.0044 \text{ mm}^2$.

TABLE II
ESTIMATED AREA PER BLOCK (SINGLE CHANNEL UNLESS NOTED)

Block	Area (μm^2)
DAC (9 channels)	3008.85
Bootstrapped Switch (9 ch.)	1362.15
StrongArm Comparator (9 ch.)	17.91
SAR Logic (9 ch.)	39.49
CLK Generator (1 $\times$)	2.15
MUX / TX Gates (9 ch.)	16.16
Total	4446.7

D. SAR Control Logic

The SAR control logic was implemented as a 9-state FSM using VerilogA (see Appendix A for the symbol). It takes in

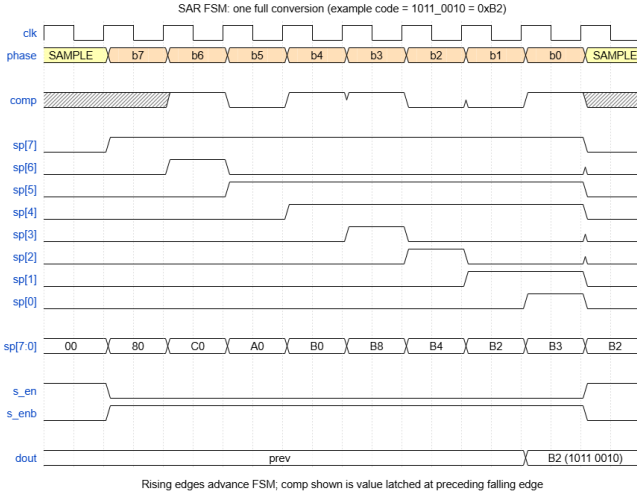


Fig. 3. Single-channel SAR ADC timing diagram for one full conversion.

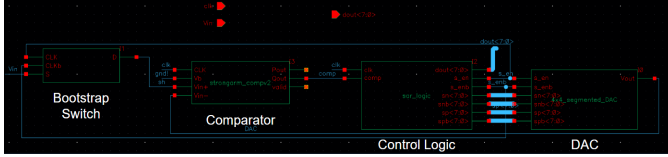


Fig. 4. Single-channel SAR ADC schematic.

the comparator decision and drives the CDAC control signals (sp/spb , sn/snb), sample enable (s_en/s_enb), and the output word $Dout$. VerilogA was chosen over a full transistor-level implementation to reduce simulation time while still accurately capturing the SAR control timing and functionality, allowing for a complete verification of time-interleaved SAR ADC behavior.

IV. SYSTEM INTEGRATION

A. Single-Channel ADC

The four sub-blocks are integrated into a complete single-channel SAR ADC (Fig. 4). The bootstrapped switch captures the input onto the sample capacitor; de-asserting s_en begins the hold phase. The comparator resolves the sign of the voltage difference on each clock edge, feeding the decision back to the FSM, which drives the DAC toward convergence bit by bit. The timing diagram in Fig. 3 shows sequencing of the comparator output, $sp[7:0]$, and s_en signals across one full 8-bit conversion.

B. $9\times$ Time-Interleaved System

Nine identical channels are driven by a 2 GHz master clock and are timed to start conversions one after another in sequence (Fig. 5). A VerilogA clock generator handles the timing by triggering each sub-channel to begin its conversion immediately after the previous one starts. An FSM then controls which transmission gate is open based on when the reset signal is triggered for the main $Dout$ output, routing the analog input

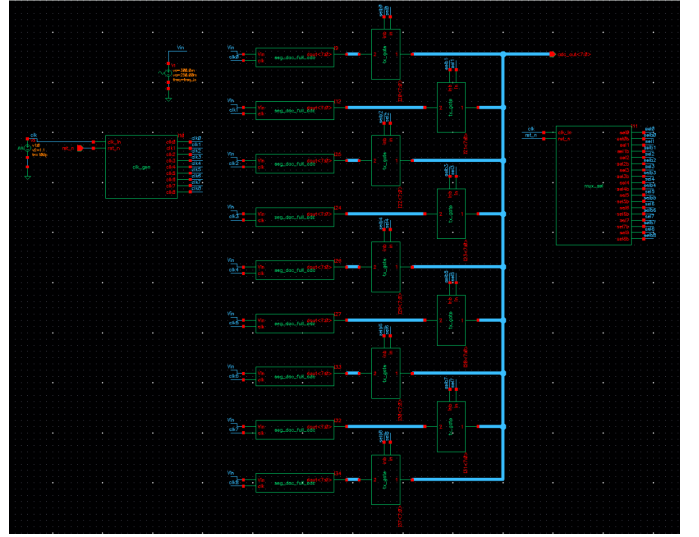


Fig. 5. Top-level schematic of the $9\times$ time-interleaved SAR ADC system.

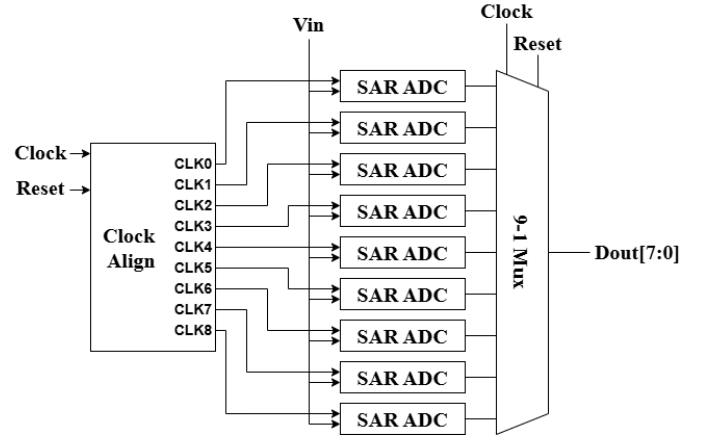


Fig. 6. Top-level block diagram of single channel SAR ADC with time-interleaving.

to the correct sub-channel's bootstrapped switch at the right time.

V. RESULTS

A. Methodology

Sinusoidal inputs were applied at 5 MHz, 990 MHz, and 1990 MHz. For each frequency, a transient simulation captured the quantized output. An FFT using 1024 samples with a Blackman window was applied. ENOB, SINAD, SNR, and SFDR were read directly from Cadence's output measurement window. Power was measured by recording I_{avg} from each supply rail (V_{DD} , V_{refp} , V_{refn} , V_{cm}) and computing total power as $P = \sum V_k \cdot I_{avg,k}$ for each of the three test frequencies; the reported power is the average across all three measurements. FOM is defined as:

$$FOM = \frac{P}{2^{ENOB} \cdot f_s}$$

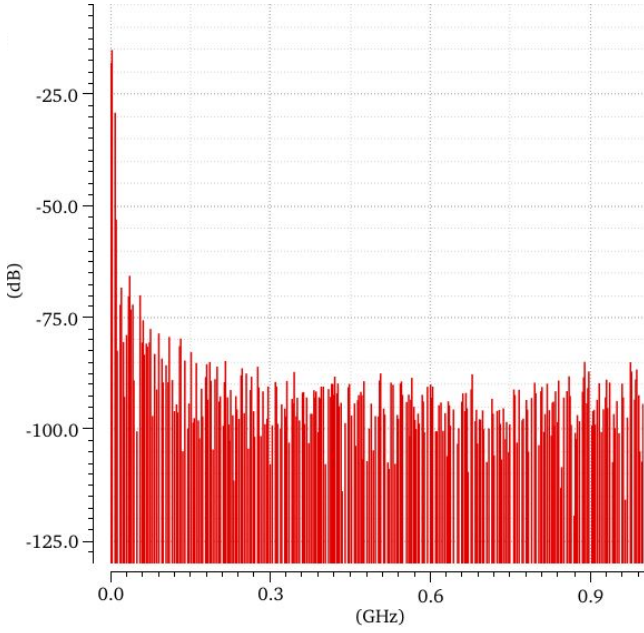


Fig. 7. FFT spectrum at 5 MHz input. The fundamental tone is visible near DC with the noise floor around -100 dBc across the Nyquist band.

B. Low-Frequency Response (5 MHz)

At a 5 MHz input (Fig. 7), the ADC produces a clean spectral output with a dominant fundamental tone and a relatively flat noise floor. The Cadence output measurements yield ENOB = 5.64 bits, SINAD = 35.74 dB, SNR = 35.74 dB, and SFDR = 46.61 dBc. The SINAD equals the SNR, indicating that at this frequency distortion and noise contribute roughly equally and no single harmonic spur dominates. The noise floor per bin is -77.40 dBc and the signal power is -14.59 dB.

C. Near-Nyquist Response (990 MHz)

At 990 MHz (Fig. 8), the fundamental tone sits near the upper edge of the Nyquist band, with its alias visible near 1 GHz. Performance degrades slightly relative to the low-frequency case: ENOB = 5.58 bits, SINAD = 35.33 dB, SNR = 35.33 dB, and SFDR = 42.23 dBc. The reduction in SFDR relative to 5 MHz is consistent with increased harmonic distortion driven by the bootstrapped switch's nonlinearity at high input slew rates and tighter settling margins at the sub-channel clock rate. The noise floor per bin is -74.49 dBc. The DAC output is shown in Figure 9 with the sample cap waveform in green and the dac output waveform in red for a 990 MHz sine wave.

D. Near-Sampling-Rate Response (1990 MHz)

At 1990 MHz (Fig. 10), the input exceeds the Nyquist limit of 1 GHz and aliases to approximately 9.77 MHz in the output spectrum, confirming correct folding behavior consistent with the 2-GSPS sampling rate. Performance degrades further as expected: ENOB = 4.18 bits, SINAD = 26.93 dB, SNR = 26.93 dB, and SFDR = 32.51 dBc. The elevated noise floor per bin (-66.27 dBc) and reduced SFDR reflect the

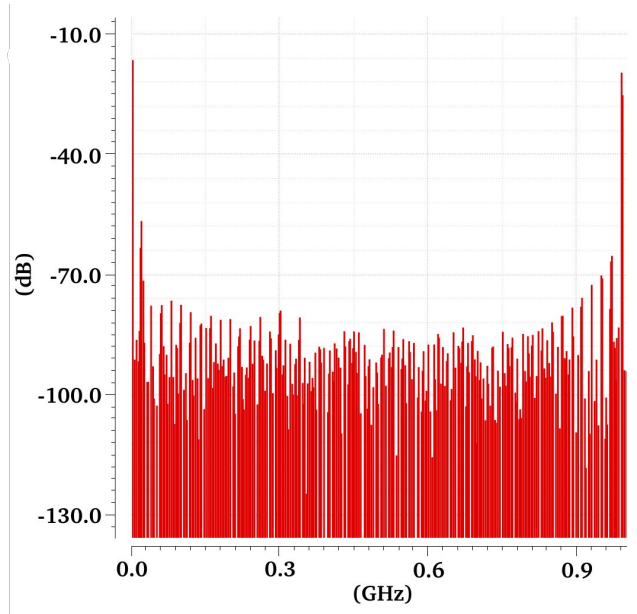


Fig. 8. FFT spectrum at 990 MHz input (near Nyquist). The fundamental appears near 0 GHz and its image appears near 1 GHz due to aliasing within the Nyquist band.

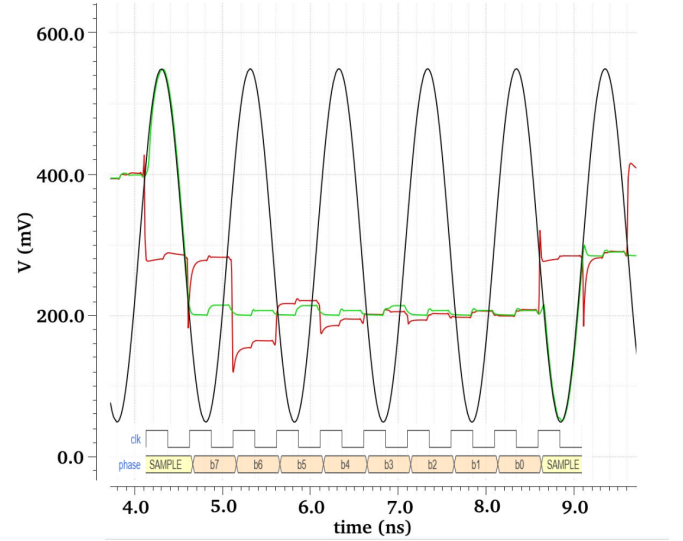


Fig. 9. SAR ADC Output Transient.

combined effect of inter-channel timing skew spurs and aliased harmonic content at this out-of-band input frequency.

E. Performance Summary

As shown in Table III, ENOB and SFDR both degrade with increasing input frequency, consistent with expected ADC behavior as the input approaches and exceeds the Nyquist limit. Notably, SNR equals SINAD at all three frequencies, indicating that no single harmonic spur dominates and that noise and distortion are comparable contributors to performance degradation. The average FOM of 46.39 fJ/step is computed

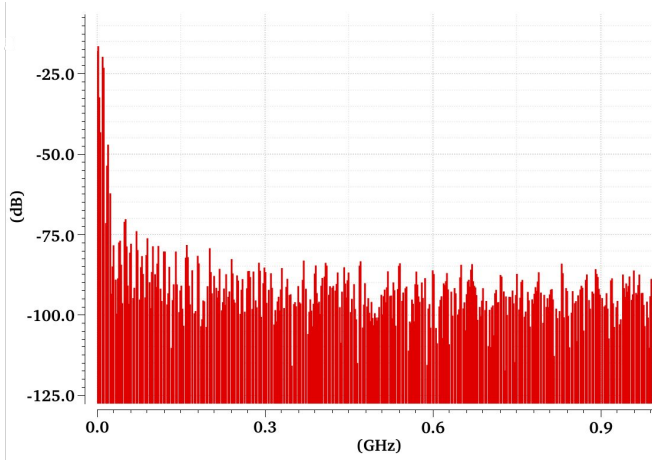


Fig. 10. FFT spectrum at 990 MHz input. The out-of-band signal aliases to ≈ 9.77 MHz in the output spectrum, confirming correct Nyquist folding behavior.

TABLE III
ADC PERFORMANCE SUMMARY

Metric	5 MHz	990 MHz	1990 MHz	Unit
SFDR	46.61	42.23	32.51	dBc
SNR	35.74	35.33	26.93	dB
SINAD	35.74	35.33	26.93	dB
ENOB	5.64	5.58	4.18	bits
Noise Floor/Bin	-77.40	-74.49	-66.27	dBc
Signal Power	-14.59	-12.08	-12.25	dB
Power	3.029	3.053	2.787	mW
Avg Power	2.956			mW
FOM	30.37	31.91	76.89	fJ/step
Avg FOM	46.39			fJ/step
Area (est.)	$4446.7 \mu\text{m}^2 \approx 0.0044 \text{ mm}^2$			—

from all three measurements using the conversion rate of 1.6×10^{10} steps/s.

F. Comparison with State-of-the-Art

TABLE IV
COMPARISON WITH PUBLISHED TI-SAR ADCs

Work	Process	fs (GS/s)	ENOB (bits)	Power (mW)	FOM (fJ/step)
[5]	55-nm	2.6	5.85	60	348
[6]	28-nm	10	5.80	21	37
[7]	28-nm	14	5.90	76.24	39
This work	45-nm	2.0	5.61	3.04	46

Table IV compares this design against published 8-bit TI-SAR ADCs. The 46 fJ/step FOM is better than the comparable-speed 55-nm design [5] but worse than the 28-nm designs [6], [7] that use asynchronous logic, monotonic switching, and digital calibration. These gaps directly motivate the future work items in Section VI. Since portions of the digital control logic are implemented in VerilogA, the reported power values underestimate the total power consumption by excluding these blocks. These blocks are expected to consume non-negligible power and may contribute significantly relative to the rest of the design.

VI. LIMITATIONS AND FUTURE WORK

Monotonic switching: The current CDAC uses a conventional switching scheme. Replacing this with a monotonic switching scheme [2] would reduce the average number of capacitor switching events per conversion cycle, directly lowering switching power and improving FOM.

Asynchronous SAR logic: The current FSM is synchronous, meaning each bit trial must wait for the next clock edge regardless of how quickly the comparator resolves. Switching to an asynchronous implementation would allow each comparison to trigger the next immediately after resolution, increasing conversion speed or reducing power at the same throughput.

Non-ideality modeling: The current design does not model several real-world effects that would impact performance in silicon. These include clock jitter, transistor and capacitor matching variations (characterized by the A_{vt} , A_C , and A_R parameters given in the design spec), power supply noise, and process-voltage-temperature (PVT) variation. Running Monte Carlo and corner simulations would give a more realistic picture of expected yield and performance across process variation.

Transistor-level SAR logic: The SAR control logic was implemented in VerilogA to simplify simulation and reduce runtime. A transistor-level implementation would capture gate delays, short-circuit currents, and layout parasitics that the behavioral model misses, giving more accurate timing and power results.

Online calibration: Because this design uses time-interleaving, any offset, gain, or timing skew between the 9 sub-channels will introduce spurious tones in the output spectrum at multiples of $f_s/9$. Implementing a digital background calibration scheme [8] would suppress these tones and recover SFDR, which is especially important for near-Nyquist performance.

VII. CONCLUSION

This paper presented the design and simulation-based verification of an 8-bit, 2-GSPS SAR ADC in a $9 \times$ time-interleaved 45-nm CMOS array with a 1.1 V supply. Simulation results show functional performance at low frequency (ENOB = 5.64 bits, SFDR = 46.61 dBc) and near-Nyquist (ENOB = 5.58 bits, SFDR = 42.23 dBc), with an average FOM of 46.39 fJ/step across all three test conditions. At 1990 MHz, the input correctly aliases to ≈ 9.77 MHz, confirming expected Nyquist folding behavior, and SNR equals SINAD across all frequencies, indicating noise and distortion contribute roughly equally. Future work will focus on monotonic CDAC switching, asynchronous SAR logic, transistor-level FSM implementation, Monte Carlo non-ideality modeling, and online calibration to correct for timing mismatch between channels.

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AI Usage Disclosure

Claude AI was used to assist in drafting portions of this report based on simulation data, refining technical phrasing, and formatting tables. All use of AI tools was conducted in accordance with the course policy as outlined in the syllabus.

APPENDIX

This appendix contains the detailed schematics and circuit symbols for the bootstrapped switch, PMOS-input StrongArm comparator, segmented CDAC, and SAR FSM sub-blocks referenced throughout the main text.

A. Bootstrapped Switch

B. PMOS-Input StrongArm Comparator

C. Segmented Capacitive DAC

D. VerilogA SAR FSM

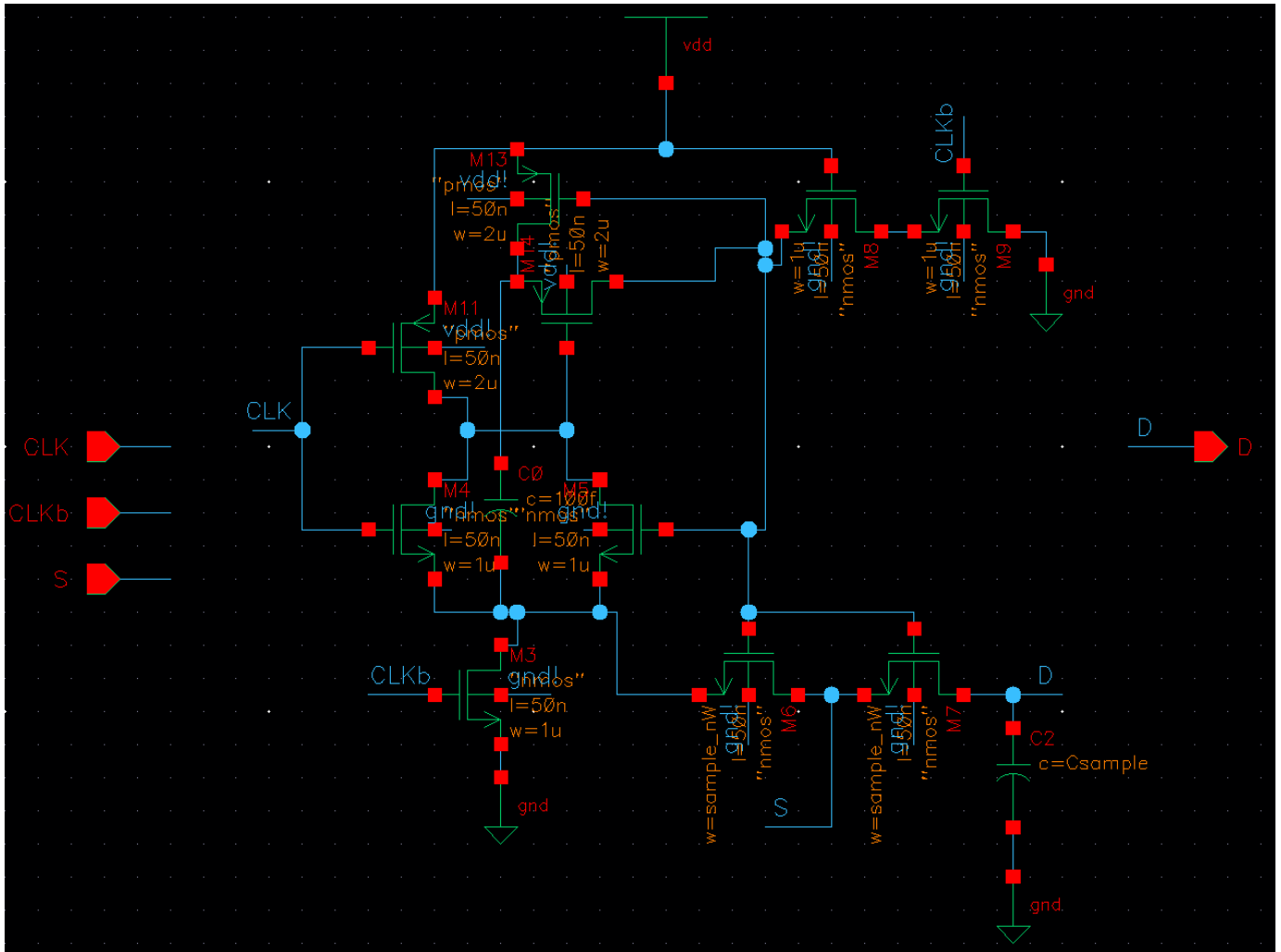


Fig. 11. Bootstrapped switch schematic.

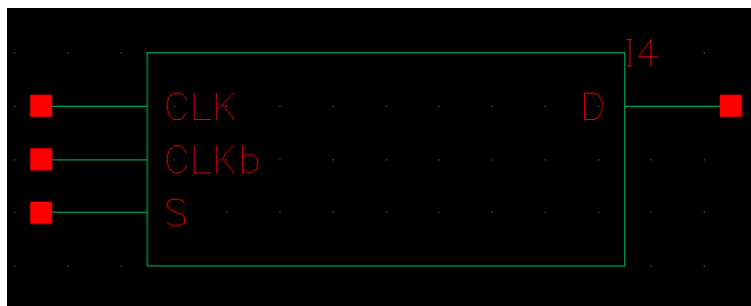
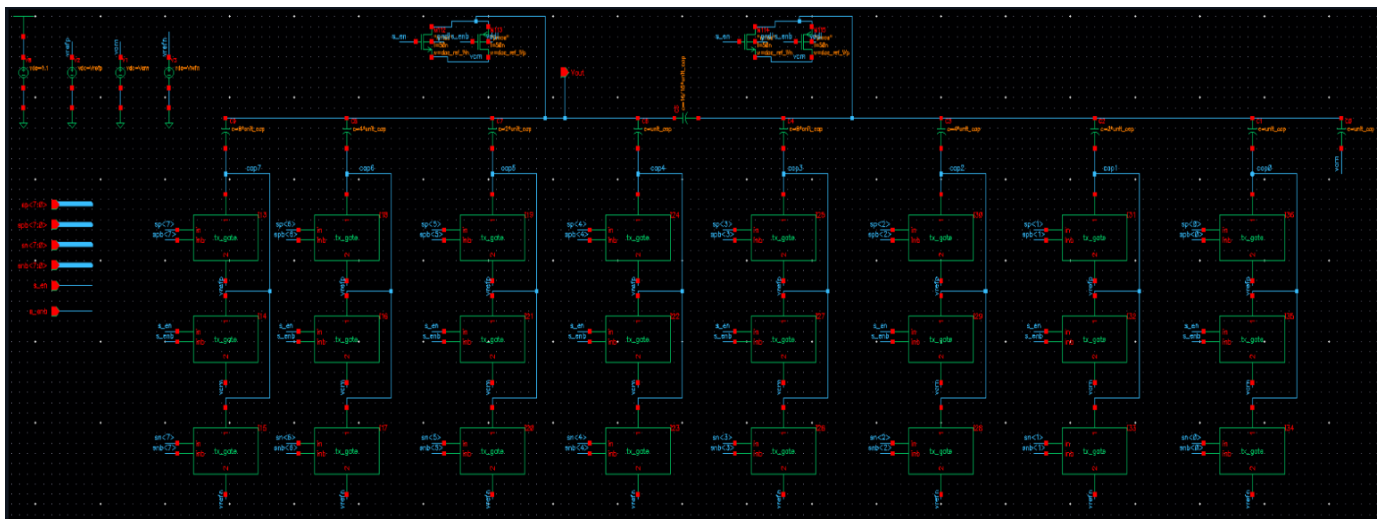
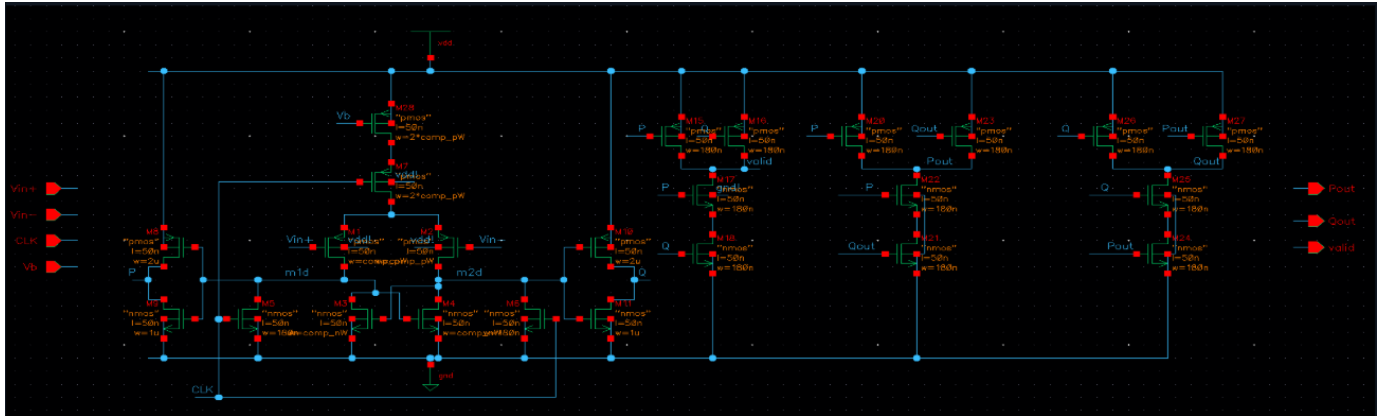


Fig. 12. Bootstrapped switch symbol.



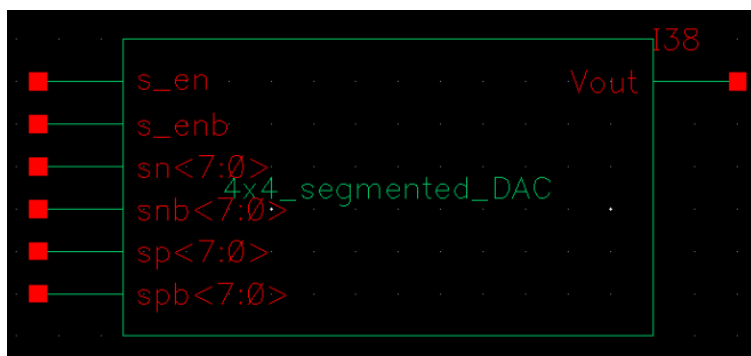


Fig. 16. Segmented CDAC symbol.

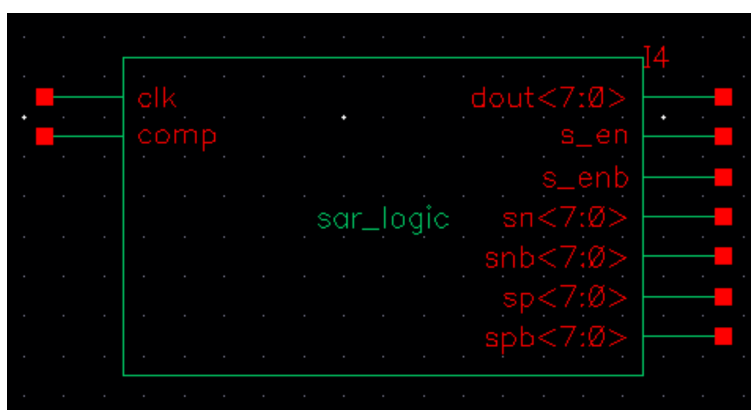


Fig. 17. VerilogA SAR FSM symbol.